

Vol 13 Chapter 2 — Genetic Code and BST Primaries

Keeper (author pass — deep math/physics revision)

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Chapter 2 — Genetic Code and BST Primaries

Level 1 — one sentence

The genetic code — 4 nucleotides (A, T, G, C) encoding 64 codons mapping to 20 amino acids — is a substrate-natural information code whose dimensions $4 = N_c + 1$, $64 = 4^{N_c}$, and $20 = C_2 \cdot N_c + 2$ are BST-primary expressions, with the codon-degeneracy reduction $64 \rightarrow 20$ as substrate Reed-Solomon compression on $GF(2^g)$.

Level 2 — graduate-physicist precision

2.1 The genetic code

DNA/RNA bases: A (adenine), T or U (thymine/uracil), G (guanine), C (cytosine). 4 bases.

Codons: triplets of nucleotides. $4^3 = 64$ codons.

Amino acids: 20 canonical (plus 3 stop signals — UAA, UAG, UGA).

The code is nearly universal across all life. Specific codon-amino acid mappings: - Methionine (start): AUG (1 codon) - Leucine: 6 codons (CUU, CUC, CUA, CUG, UUA, UUG) — high degeneracy - Tryptophan, methionine: 1 codon each — low degeneracy

Degeneracy concentrated in 3rd position (wobble).

2.2 BST-primary expressions

- 4 nucleotides: $N_c + 1 = 4$
- 64 codons: $4^{N_c} = 4^3 = 64$
- 20 amino acids (S-tier structural numerology, demoted per Grace INV-5113 sweep 2026-05-24): Grace's exhaustive 3-primary search found 53 distinct BST-primary expressions equal to 20 from 10 primaries (null-model rate ~5%, well above the Cal #44 risk threshold). Examples: $C_2 \cdot N_c + 2 = 20$, $N_c + \text{seesaw} = 3 + 17 = 20$, $g + c_3 = 7 + 13 = 20$, $\chi + \text{rank} - C_2 = 24 - 4 = 20$, ..., plus 48 more. No expression is privileged without a mechanism-forcing derivation. Promotion to I-tier+ requires Lyra task #306 to produce the substrate-dynamics derivation that uniquely picks one expression.
- 23 (20 + 3 stops): cf. $g \cdot N_c + 2 = 23$, or M_24 group order = 24 with one orbit missing
- 3 in "3-nucleotide codon": N_c literally

2.3 Substrate compression view

Codons (64 states) compress to amino acids (20 states) + stops (3 states): factor ~ 3 redundancy.

This is Reed-Solomon-like coding (Vol 14 Ch 9): some redundancy in codon space buys robustness against transcription errors. Wobble base pairing (third position degeneracy) IS the redundancy mechanism.

Substrate view (Paper #122): genetic code is biology's instance of substrate Reed-Solomon coding on $GF(2^g) = GF(128)$, with codon $\leftrightarrow$ amino acid as substrate symbol $\rightarrow$ message-symbol compression.

2.4 Universality

Same code across bacteria, archaea, eukaryotes. Minor variants (mitochondrial code, mycoplasma, ciliates) differ in 1-3 codons.

Universality argues for single common ancestor (LUCA) or substrate-forced. BST reads it as substrate-forced: the BST-primary dimensions are the natural code.

2.5 K-audit anchors

- Vol 14 Ch 9 (Reed-Solomon on GF(128) substrate code)
- Paper #122 (Information substrate)
- K59 cyclotomic (substrate code framework)

Level 3 — 5th-grader accessibility

Genetic code: DNA has 4 letters (A, T, G, C). Reads in 3-letter words (codons): $4^3 = 64$. Codons code for 20 amino acids (plus 3 “stops”). In BST: $4 = N_c + 1$, $64 = 4^{N_c}$, $20 = S$ -tier structural numerology (53 BST-primary expressions match — no mechanism-forced canonical yet per Grace INV-5113) — first two are substrate-natural, “20” awaits mechanism. Degeneracy (multiple codons → same amino acid) is Reed-Solomon error-correction: the substrate’s universal coding mechanism applied at the genetic scale.

What comes next

Chapter 3 develops prebiotic forcing.

Where to look this up

- Crick, Watson 1953 (DNA structure)
- BST: Vol 14 Ch 9; Paper #122